How Close Are We to AGI?

A Reality Check for Decision-Makers

ETH Zurich Executive Education

2026

Session Overview — Five Acts in Ninety Minutes

Act	Theme	Anchor reference
1	What IS AGI?	Morris et al., ICML 2024
2	Sparks & Specialist Breakthroughs	Bubeck 2023; *Nature* 2024
3	Chatbots to Agents	METR, 2024–2026
4	When Will AGI Arrive?	Grace et al. 2024, N=2778
5	Managing the Transition	Bengio–Hinton–Russell, *Science* 2024

Act 1 — What IS AGI?

Act 1

What Do We Mean by AGI?

Anchor: Morris et al., "Levels of AGI," ICML 2024

The Question Everyone Is Asking

"Is this thing actually intelligent — or just very good at autocomplete?"

- The phrase "AGI" is now in press releases, contracts, and regulation.
- Five years ago, serious researchers avoided it in public.
- **If the term is fuzzy, the obligation is fuzzy.**

The Morris et al. Framework — Two Axes

	Narrow	General
Superhuman	AlphaFold, AlphaZero	*not yet*
Virtuoso (99th %)	Deep Blue	*not yet*
Expert (90th %)	grammar checkers	*not yet*
Competent (50th %	Siri-era assistants	*not yet*
Emerging	early rule syster	**GPT-4 / Gemini / Claude**

Adapted from Morris et al. (2024).

Where Frontier Models Sit Today

"Emerging General AGI."

- **More than** the public sometimes hears: a genuinely new category of artefact.
- **Less than** the marketing sometimes claims: not yet *competent* across domains.
- **No press release** will announce AGI. You will notice it workflow by workflow.

Counterweight: Chollet's "Measure of Intelligence"

*Intelligence is the **efficiency** with which a system acquires new skills in a previously unseen domain, using limited prior knowledge and limited experience.*

- Benchmark scores can reflect *memorisation* + *interpolation*.
- ARC-AGI: designed to resist training-data overlap.
- Was only substantially cracked in late 2024 — and at high compute cost.

Chollet, 2019

Act 1 Takeaway

- We have a vocabulary: **performance** $\times$ **generality**.
- Today's frontier models: **Emerging General AGI**.
- When a vendor says "approaching AGI," ask:

  On which axis? Against what reference distribution?

Act 2 — Sparks & Specialist Breakthroughs

Act 2

What Can the Best Systems Already Do?

Anchors: Bubeck et al., Sparks of AGI, 2023;

Trinh et al., AlphaGeometry, Nature 2024

Bubeck et al., “Sparks of AGI”

“We could reasonably view GPT-4 as an early (yet still incomplete) version of an AGI system.”

- 155-page technical report, Microsoft Research, March 2023.
- Qualitative probes: maths, code, vision-via-TikZ, theory of mind, analogy.
- The *claim* broke the taboo. The *paper* is honest about its limits.

Bubeck et al., 2023

Sparks — The Honest Caveats

- Tested an **early, pre-alignment** GPT-4 (no RLHF yet).
- **Could not inspect training data** — contamination possible.
- Authors explicit about **cherry-picking interesting examples**.
- Schaeffer–Miranda–Koyejo (NeurIPS 2023, Outstanding Paper): "emergence" partly a *measurement* artefact.

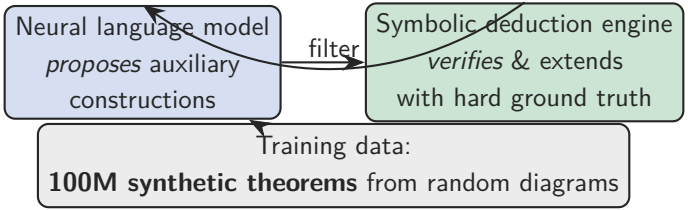
Schaeffer et al., 2023

AlphaGeometry — The Problem

- IMO-level Euclidean geometry: *creative* insight required.
- Typical trick: introduce an auxiliary point, line, or circle.
- After the trick: symbolic deduction finishes the proof.
- Human geometry proofs are **scarce and hard to formalise**.

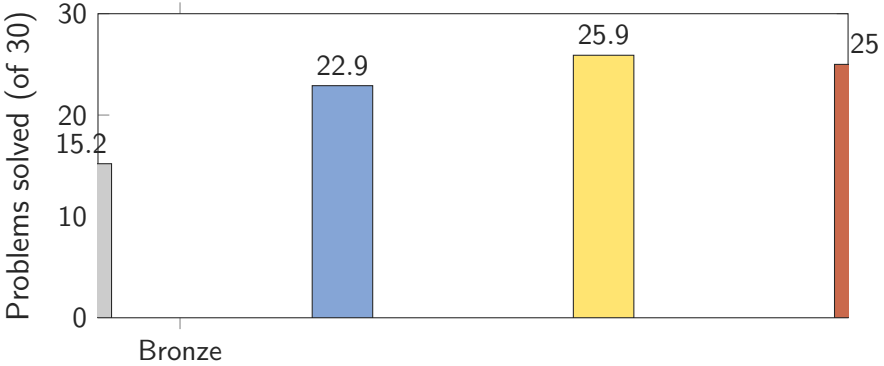
Trinh et al., 2024

AlphaGeometry — Neuro-Symbolic Architecture



Zero human-written proofs in training.

AlphaGeometry — 25 out of 30 IMO Problems



Human medallist averages vs. AlphaGeometry, 30 recent IMO geometry problems.

The Pattern That Matters for Managers

"Domains with verifiers fall fast.
Domains without verifiers do not."

- Verifier-rich: maths, code, games, formal compliance checks.
- Verifier-poor: taste, contested judgement, strategic persuasion.
- **Which parts of your business have a verifier — or could?**

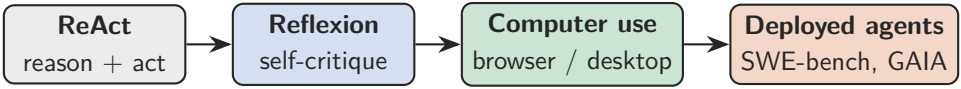
Act 3 — From Chatbots to Agents

Act 3

From Chatbots to Agents: Measuring Autonomy

Anchor: METR, "Measuring AI Ability to Complete Long Tasks," 2024–2026

What Is an "Agent"?



Unit of work: one-shot output $\longrightarrow$ multi-stage trajectory.

Implication: the right question becomes *"for how long?"*

Yao et al., 2023; Shinn et al., 2023

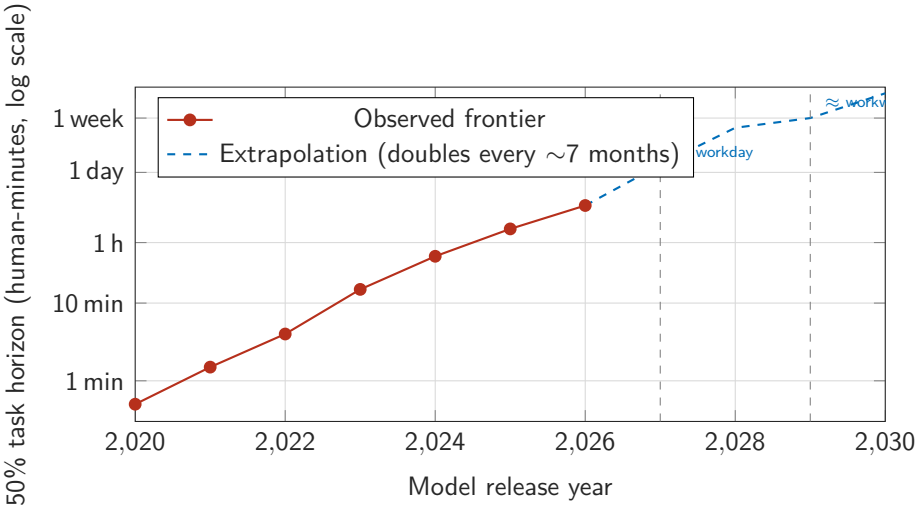
METR — The Task-Horizon Metric

- Portfolio of real engineering, research, data-analysis tasks.
- Measure **how long a skilled human needs** for each one.
- Ask: at what human-time horizon does the model succeed $\geq 50\%$ of the time?
- That number is the **50% task horizon**.

Crucially: third-party, not a vendor benchmark; rigorously human-baselined; tracks a trend rather than saturating.

METR, 2024–2026

The METR Doubling Curve



Extrapolation Is a Claim, Not a Measurement

Reasons the curve could bend down:

- Compute cost; energy and chip constraints.
- Data exhaustion for next-token pre-training.
- Long-horizon reliability is *engineering*, not just scaling.
- A capability ceiling we do not yet understand.

Reasons it could accelerate:

- Reasoning models; better post-training; tool use; multi-agent systems.

Track the curve quarterly. Re-plan when it bends.

Supporting Evidence — Same Shape, Different Axes

- **SWE-bench** (Jimenez et al., 2024) — real GitHub issues. From unusable to $> 50\%$ on Verified variants.
- **GAIA** (Mialon et al., 2024) — general assistant tasks. Humans $\sim 92\%$; frontier systems from $< 15\%$ to $> 50\%$ in 2.5 years.
- Both are contamination-resistant by construction.
- Both trace the same curve as the METR horizon.

Act 4 — When Will AGI Arrive?

Act 4

When Will AGI Arrive?

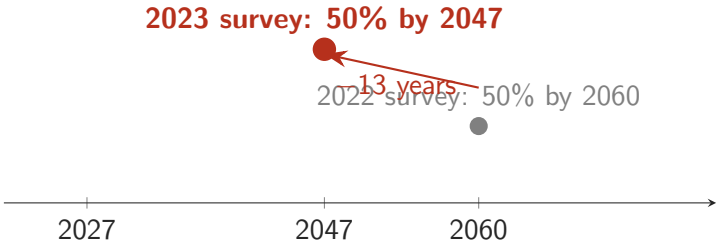
Anchor: Grace et al., Thousands of AI Authors on the Future of AI, 2024

The Largest Expert Survey Ever Conducted

- **2,778 researchers** who published at NeurIPS / ICML.
- Survey run January 2023; prior run in 2022.
- Asks: when will *High-Level Machine Intelligence* (HLMI) — machines better and cheaper than humans at every task — arrive?
- Also: specific-capability timelines, and probabilities of extreme outcomes.

Grace et al., 2024

The 13-Year Shift



Same respondents. One year apart. Updated 13 years earlier.

10% probability date: 2027. 90% date: >2100.

The Disagreement *Is* the Signal

- Aggregates hide a community split by **fifty years**.
- A serious minority expects HLMI before 2030.
- Another serious minority expects it after 2100.
- **38% give** $\geq$ 10% **probability to** "extremely bad" outcomes.

A 10% tail risk stops a drug trial. Should it stop anything here?

Act 5 — Managing the Transition

Act 5

Managing the Transition

Anchor: Bengio, Hinton, Russell et al.,
"Managing extreme AI risks amid rapid progress,"
Science 384, 842–845 (2024)

The *Science* Consensus Statement

Signatories include:

- **Yoshua Bengio** — Turing Award
- **Geoffrey Hinton** — Turing Award, 2024 Nobel in Physics
- **Stuart Russell** — author of the standard AI textbook
- **Andrew Yao** — Turing Award
- **Daniel Kahneman** — Nobel in Economics
- **Yuval Noah Harari**, Dawn Song, Pieter Abbeel, Trevor Darrell, . . .

Published in *Science*, not a blog post. Not a policy white paper. Not a letter.

Bengio et al., 2024

The Four Governance Asks

1. **Reorient R&D.** Serious fraction of effort to safety & ethics (paper suggests $\sim 1/3$).
2. **Adaptive governance.** Pre-deployment evals, incident reporting, ability to pause.
3. **Commensurate risk management.** Aviation-, pharma-, nuclear-style standards; third-party audits; liability.
4. **International coordination.** Aligned standards, with compliance verification.

These are converging with EU AI Act, UK/US AISI evaluations, Bletchley & Seoul declarations.

Manager Playbook — Monday Morning

1. **Pilot** a frontier model on 1–2 *named* workflows with named owners.
2. **Build** an internal evaluation function (start: one person).
3. **Track** three external signals quarterly:
   METR horizon · SWE-bench Verified / GAIA · UK AISI / US AISI.
4. **Include** AI risk in your enterprise risk review cycle.
5. **Avoid** irreversible commitments: multi-year exclusives, unrevertible workflows, lock-in on proprietary evals.

The One Thing To Remember

"The question is not whether AGI is coming.

The question is whether your organisation has the

evaluation capability to tell — month by month,

workflow by workflow — what these systems

can and cannot do for you."

References

Yoshua Bengio, Geoffrey Hinton, Andrew Yao, Dawn Song, Pieter Abbeel, Trevor Darrell, Yuval Noah Harari, Ya-Qin Zhang, Lan Xue, Shai Shalev-Shwartz, Gillian Hadfield, Jeff Clune, Tegan Maharaj, Frank Hutter, Atılım Güneş Baydin, Sheila McIlraith, Qiqi Gao, Ashwin Acharya, David Krueger, Anca Dragan, Philip Torr, Stuart Russell, Daniel Kahneman, Jan Brauner, and Sören Mindermann. Managing extreme AI risks amid rapid progress. *Science*, 384 (6698):842–845, 2024. doi: 10.1126/science.adn0117.

Reading List ii

Sébastien Bubeck, Varun Chandrasekaran, Ronen Eldan, Johannes Gehrke, Eric Horvitz, Ece Kamar, Peter Lee, Yin Tat Lee, Yuanzhi Li, Scott Lundberg, Harsha Nori, Hamid Palangi, Marco Tulio Ribeiro, and Yi Zhang. Sparks of artificial general intelligence: Early experiments with GPT-4. *arXiv preprint arXiv:2303.12712*, 2023. URL `https://arxiv.org/abs/2303.12712`.

François Chollet. On the measure of intelligence. *arXiv preprint arXiv:1911.01547*, 2019.

Epoch AI. Tracking compute trends across frontier AI models. Technical report, Epoch AI, 2024–2026. epochai.org.

Reading List iii

Katja Grace, Harlan Stewart, Julia Fabienne Sandkühler, Stephen Thomas, Ben Weinstein-Raun, and Jan Brauner. Thousands of AI authors on the future of AI. *arXiv preprint arXiv:2401.02843*, 2024. URL `https://arxiv.org/abs/2401.02843`.

Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? In *International Conference on Learning Representations (ICLR)*, 2024.

Reading List iv

Nestor Maslej, Loredana Fattorini, Ray Perrault, et al. Artificial intelligence index report 2024. Technical report, Stanford Institute for Human-Centered AI (HAI), 2024.

METR. Measuring AI ability to complete long tasks. Model Evaluation and Threat Research, technical reports, 2024–2026. Task-horizon doubling-time analyses; see metr.org/blog.

Grégoire Mialon, Clémentine Fourrier, Craig Swift, Thomas Wolf, Yann LeCun, and Thomas Scialom. GAIA: A benchmark for general AI assistants. In *International Conference on Learning Representations (ICLR)*, 2024.

Reading List v

Meredith Ringel Morris, Jascha Sohl-Dickstein, Noah Fiedel, Tris Warkentin, Allan Dafoe, Aleksandra Faust, Clément Farabet, and Shane Legg. Levels of AGI for operationalizing progress on the path to AGI. *arXiv preprint arXiv:2311.02462*, 2024. URL `https://arxiv.org/abs/2311.02462`. Published at ICML 2024 as "Position: Levels of AGI for Operationalizing Progress on the Path to AGI".

Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language models a mirage? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. Outstanding Paper Award.

Reading List vi

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Trieu H. Trinh, Yuhuai Wu, Quoc V. Le, He He, and Thang Luong. Solving olympiad geometry without human demonstrations. *Nature*, 625(7995): 476–482, 2024. doi: 10.1038/s41586-023-06747-5.

UK AI Safety Institute. AI safety institute approach to evaluations. Technical report, UK AI Safety Institute, 2024. aisi.gov.uk.

Reading List vii

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

Thank you

Questions?

Session companion materials and annotated bibliography available in the course repository.